

Week 4 Recitation

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ECON8711 (Micro 1A)

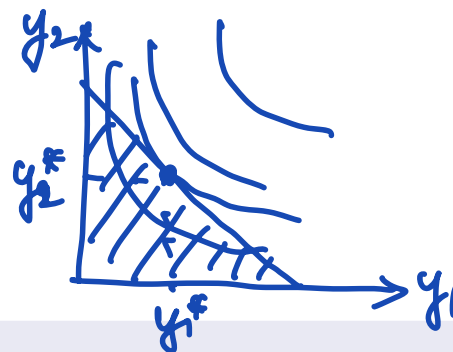
Sep 15, 2023

Please let me know your thoughts and suggestions
till every recitation class:

<https://forms.gle/jm3d4iJ51tjUnTiD9>

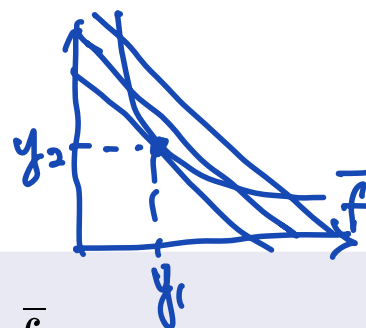
Duality

Utility (Preference) Max



$$\max_{y \in \mathbb{R}^N} f(y) \text{ s.t. } \underline{p \cdot y \leq m.}$$

Expenditure Min



$$\min_{y \in \mathbb{R}^N} p \cdot y \text{ s.t. } f(y) \geq \bar{f}.$$

Duality

- i) $\underline{p' \cdot \hat{y}} < 1$
 $\exists p'' \text{ s.t. } p'' \cdot \hat{y} \leq 1 \text{ \& } \underline{V(p'')} < \underline{V(p')}$
 ii) $p' \cdot \hat{y} = 1 \text{ (} p' \neq \hat{p} \text{)}$

dual ②

$$\underline{h(y^*(p))} = \min_{\substack{p \in \mathbb{R}^N \\ \hat{p}}} \underline{V(p)} \text{ s.t. } \underline{p \cdot y} \leq 1$$

Claim) $\underline{V(p')} > \underline{V(\hat{p})}$
 $= \max_y \underline{f(y)} \text{ s.t. } \underline{p' \cdot y} = 1$

$$\underline{e(p, m)} = \min_{\substack{y \in \mathbb{R}^N \\ \underline{m}}} p \cdot y \text{ s.t. } f(y) \geq \underline{\bar{f}}.$$

$\underline{\bar{f}(m)}$

Envelop Theorem

Consider the following problem: $\theta = (p, m)$

$$\checkmark \max_{y \in \mathbb{R}^N} \underline{f(y, \theta)} \text{ s.t. } \underline{g(y, \theta)} \leq 0$$

Envelope Theorem:

Assume that

f, g are C^1 functions in $W \times U$,

and for $\forall \theta \in U$, $\exists! y(\theta) \in W$ that is C^1 , $g(y(\theta), \theta) \geq 0$

Let

- $\mathcal{L}(y, \theta, \lambda)$ be the Lagrangian function,
- $y(\theta), \lambda(\theta)$ solve the KT conditions,
- $V(\theta)$ be the value function

Then $\forall \theta \in U$,

$$\frac{\partial V(\theta)}{\partial \theta} = \frac{\partial \mathcal{L}(y, \theta, \lambda)}{\partial \theta} \bigg|_{y(\theta), \lambda(\theta)}$$

Envelope Theorem

$$L(y_1, y_2; p_1, p_2, m) = \max_{y_1, y_2} f(y_1, y_2) + \lambda(m - p_1 y_1 - p_2 y_2)$$

$$\frac{\partial L}{\partial m} = \lambda^* \Rightarrow \frac{\partial V(p_1, p_2, m)}{\partial m} = \frac{\partial L}{\partial m} \Big|_{\lambda^*, y^*}$$

Why is this useful?:

I want to know what is the effect of a parameter on the welfare of a decision maker

$$\frac{\partial L}{\partial p_1} = -\lambda^* y_1^*$$

$$y_1^* = - \frac{\frac{\partial V}{\partial p_1}}{\frac{\partial V}{\partial m}}$$

Roy's Identity (Recovering demand)

$$\left[- \frac{\frac{\partial V(p_1, p_2, m)}{\partial p_1}}{\frac{\partial V(p_1, p_2, m)}{\partial m}} \right] = y_1(p_1, p_2, m)$$

Some Qual-type questions on Duality

(Leontief) $\max_{x_1, x_2, x_3 \geq 0} \left(\underbrace{x_1^{\frac{1}{2}}}_{\text{S.t.}} + \underbrace{\min \{ \ddot{x}_2, \ddot{x}_3 \}}_y \right) x$

$p_1 = p_x$
 $p_2 + p_3 = p_y$
 $\ddot{x}_2 \leq x_3 \quad (i)$
 $x_2^* = y^*$
 $x_3 \geq y^*$

$p_1 x_1 + p_2 x_2 + p_3 x_3 \leq m$

$$\mathcal{L} = x^{\frac{1}{2}} + y + \lambda [m - p_x x - p_y y] + \mu_1 x + \mu_2 y \quad (\lambda, \mu_1, \mu_2 \geq 0)$$

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2x^{\frac{1}{2}}} - \lambda p_x + \mu_1 = 0$$

$$\Leftrightarrow \lambda p_x = \frac{1}{2x^{\frac{1}{2}}} \Leftrightarrow \lambda = \frac{1}{p_x} \frac{1}{2x^{\frac{1}{2}}}$$

$$\star \begin{cases} x^* > 0 \\ \mu_1^* = 0 \end{cases}$$

$$\frac{\partial \mathcal{L}}{\partial y} = 1 - \lambda p_y + \mu_2 = 0$$

case 1) $y^* = 0$ ✓
 $x^* = \frac{m}{p_x}$

($\lambda > 0$) $m - p_x x - p_y y = 0$ ✓

$$\lambda p_x = \frac{1}{2x^{\frac{1}{2}}}$$

$$\frac{p_x}{p_y} = \frac{1}{2x^{\frac{1}{2}}}$$

$$\frac{1}{4x} = \left(\frac{p_x}{p_y} \right)^2$$

$$x^* = \frac{p_y^2}{4p_x^2}$$

case 2) $y^* > 0$. $p_y y^* = m - p_x x$
 $\mu_2^* = 0$ $y^* = \frac{1}{p_y} [m - p_x x]$
 $\lambda^* = \frac{1}{p_y}$

eg) $\ln(x_1 + x_2) f_1$
 $+ \ln(x_1 + x_2 + x_3) f_2$ ←

$$V(x_1, x_2, x_3; p, m)$$

$$= \max_{x_1, x_2, x_3} L$$

where

$$L = \underline{\hspace{2cm}}$$

$$\left[\begin{array}{ccc} \frac{\partial V}{\partial m} & & \\ \frac{\partial V}{\partial p_1} & \frac{\partial V}{\partial p_2} & \frac{\partial V}{\partial p_3} \end{array} \right]$$

By the
E.T.

$$-x_i^* = - \frac{\frac{\partial V}{\partial p_i}}{\frac{\partial V}{\partial m}} \quad \checkmark$$

$$x_2^* = - \frac{\frac{\partial V}{\partial p_2}}{\frac{\partial V}{\partial m}}$$